

# Negative Exponent Law

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**MA.8.NSO.1.3**

I can rewrite a power with a negative exponent using a positive exponent, and evaluate it.

# Warm-Up

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1. Find the value:  $\frac{5^4 \cdot 5^0}{5^2}$
2. Which is greater,  $\sqrt{30}$  or 5.5?
3. Solve  $-3x = 21$
4. What is the reciprocal of  $\frac{3}{4}$ ? What is the opposite of  $\frac{3}{4}$ ?

# Warm-Up

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1. Find the value:  $\frac{5^4 \cdot 5^0}{5^2}$  **25**
2. Which is greater,  $\sqrt{30}$  or 5.5? **5.5**
3. Solve  $-3x = 21$   **$x = -7$**
4. What is the reciprocal of  $\frac{3}{4}$ ? What is the opposite of  $\frac{3}{4}$ ?  
 **$\frac{4}{3}$  and  $-\frac{3}{4}$**

# Notes

## I. The pattern keeps going past zero

| $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ | $2^{-1}$      | $2^{-2}$      | $2^{-3}$      |
|-------|-------|-------|-------|-------|---------------|---------------|---------------|
| 16    | 8     | 4     | 2     | 1     | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{8}$ |

$$2^{-1} = \frac{1}{2^1}$$

$$2^{-2} = \frac{1}{2^2} = \frac{1}{4}$$

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$$

# Notes

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## II. The negative exponent law

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$$a^{-n} = \frac{1}{a^n} \quad \frac{1}{a^{-n}} = a^n \quad (a \neq 0)$$

$$5^{-2} = \frac{1}{5^2} = \frac{1}{25} \quad \text{but} \quad -5^2 = -25 \quad \text{and} \quad (-5)^2 = 25$$

**A negative exponent means reciprocal. It never means a negative value.**

The base keeps its own sign: 
$$(-2)^{-3} = \frac{1}{(-2)^3} = \frac{1}{-8} = -\frac{1}{8}$$

# Notes

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## III. The laws still work with negative exponents

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$$6^{-2} \cdot 6^5 = \frac{1}{6^2} \cdot 6^5 = \frac{7776}{36} = 216$$

$$\text{or } 6^{-2} \cdot 6^5 = 6^{-2+5} = 6^3 = 216$$

$$\frac{9^{-5}}{9^3} = 9^{-5-3} = 9^{-8} = \frac{1}{9^8}$$

**Add, subtract and multiply exponents exactly as before — the exponents are just integers now.**

# Example 1

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$$\frac{2^{-3}}{9^{-2}}$$

**Rewrite with positive exponents, then find the value.**

# Example 1

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$$\frac{2^{-3}}{9^{-2}} = \frac{9^2}{2^3}$$

$$\frac{9^2}{2^3} = \frac{81}{8} = 10.125$$

**Answer:  $81/8 = 10.125$**

# Your Turn

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$$\frac{3^{-2}}{5^{-1}}$$

**Rewrite with positive exponents, then find the value.**

# Your Turn

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$$\frac{3^{-2}}{5^{-1}}$$

Rewrite with positive exponents, then find the value.

$$3^{-2} = \frac{1}{3^2} \quad \text{and} \quad \frac{1}{5^{-1}} = 5^1$$

$$\frac{3^{-2}}{5^{-1}} = \frac{5^1}{3^2} = \frac{5}{9}$$

**Answer:**  $\frac{5}{9}$

## Example 2

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$$\frac{7^0}{7^{-4}}$$

**Find the value.**

## Example 2

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$$\frac{7^0}{7^{-4}} = 7^{0 - (-4)} = 7^4$$

$$\frac{7^0}{7^{-4}} = \frac{1}{7^{-4}} = 7^4$$

$$7^4 = 2401$$

**Answer: 2,401**

# Your Turn

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$$\frac{(-8)^{-4}}{(-8)^{-6}}$$

**Find the value.**

# Your Turn

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$$\frac{(-8)^{-4}}{(-8)^{-6}}$$

Find the value.

$$\begin{aligned}\frac{(-8)^{-4}}{(-8)^{-6}} &= (-8)^{-4 - (-6)} = (-8)^2 \\ (-8)^2 &= 64\end{aligned}$$

**Answer: 64**

# Whiteboards · Question 1 of 9

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$$8^{-3}$$

**Answer it.**

# Whiteboards · Question 1 of 9

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$$8^{-3}$$

$$8^{-3} = \frac{1}{8^3}$$

**Answer:**  $\frac{1}{8^3}$

# Whiteboards • Question 2 of 9

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A sheet of paper 30 centimeters long is folded in half 7 times.

Each fold makes the top layer half as big.

**What fraction of the whole sheet is the top layer after 7 folds?**

**Answer it.**

# Whiteboards • Question 2 of 9

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A sheet of paper 30 centimeters long is folded in half 7 times.

Each fold makes the top layer half as big.

**What fraction of the whole sheet is the top layer after 7 folds?**

Each fold halves it:  $\left(\frac{1}{2}\right)^7 = \frac{1}{2^7}$

$$2^7 = 128, \text{ so } \frac{1}{2^7} = \frac{1}{128}$$

**Answer:**  $\frac{1}{128}$

# Whiteboards • Question 3 of 9

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$$\frac{1}{10^{-2}}$$

**Answer it.**

# Whiteboards • Question 3 of 9

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$$\frac{1}{10^{-2}}$$

$$\frac{1}{10^{-2}} = 10^2$$

$$10^2 = 100$$

**Answer:  $10^2 = 100$**

# Whiteboards · Question 4 of 9

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$$(-2)^{-3}$$

**Answer it.**

# Whiteboards • Question 4 of 9

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$$(-2)^{-3}$$

$$(-2)^{-3} = \frac{1}{(-2)^3}$$

$$(-2)^3 = -8, \text{ so } \frac{1}{(-2)^3} = -\frac{1}{8}$$

**Answer:**  $-\frac{1}{8}$

# Whiteboards • Question 5 of 9

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$$5^{-2}$$

Which is the value?

**A.** 1/10

**B.** 1/25

**C.** -25

**D.** -10

**Answer it.**

# Whiteboards • Question 5 of 9

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$$5^{-2}$$

Which is the value?

$$5^{-2} = \frac{1}{5^2} = \frac{1}{25}$$

**Answer: B — 1/25**

# Whiteboards • Question 6 of 9

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$$6^{-2} \cdot 6^5$$

Which expression is equivalent?

**A.**  $6^{-10}$

**B.**  $36^3$

**C.**  $6^{-7}$

**D.**  $6^3$

**Answer it.**

# Whiteboards • Question 6 of 9

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$$6^{-2} \cdot 6^5$$

Which expression is equivalent?

$$6^{-2} \cdot 6^5 = 6^{-2+5} = 6^3$$

**Answer: D —  $6^3$**

# Whiteboards · Question 7 of 9

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$$3^{-4} \cdot 3^4$$

**Answer it.**

# Whiteboards · Question 7 of 9

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$$3^{-4} \cdot 3^4$$

$$3^{-4} \cdot 3^4 = 3^{-4+4} = 3^0 = 1$$

**Answer: 1**

# Whiteboards • Question 8 of 9

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$$\frac{9^{-5}}{9^3}$$

**Answer it.**

# Whiteboards • Question 8 of 9

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$$\frac{9^{-5}}{9^3}$$

$$\frac{9^{-5}}{9^3} = 9^{-5-3} = 9^{-8}$$

$$9^{-8} = \frac{1}{9^8}$$

**Answer:**  $\frac{1}{9^8}$

# Whiteboards • Question 9 of 9

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Ellen says the value of  $5^{-2}$  is  $-25$ .

Explain why she is wrong, and show the correct value.

**Write your answer in sentences.**

# Whiteboards • Question 9 of 9

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Ellen says the value of  $5^{-2}$  is  $-25$ .

Explain why she is wrong, and show the correct value.

A negative exponent means the reciprocal, not the opposite.

$$5^{-2} = \frac{1}{5^2} = \frac{1}{25}$$

**Answer:  $5^{-2} = 1/5^2 = 1/25$**

# Independent Set

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1. Rewrite with a positive exponent. Do not evaluate.  $9^{-2}$
2. Find the value.  $3^{-4}$
3. Find the value.  $(-4)^{-2}$
4. Find the value.  $\frac{1}{10^{-3}}$
5. Find the value.  $5^2 \cdot 5^{-5}$
6. Find the value of  $n$ .  $2^n = \frac{1}{16}$

# IXL

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1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

## **Today's skills — all required**

Understanding negative exponents — YBB

Evaluate powers with negative exponents — WGS